

A Packing of 12 Four-Spaces in $\mathbb{R}^{12}$

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Abstract

We give 12 4-dimensional subspaces of $\mathbb{R}^{12}$ whose smallest squared chordal distance is at least 2.908757. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.888584, obtained by deleting 1 of the 13 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 12 subspaces of dimension 4 in $\mathbb{R}^{12}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.908757 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 12$, $n = 4$. The cell $N = 12$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.888584, obtained by deleting 1 of the 13 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(12 - 4)/12 \cdot 12/11 = 2.909091$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m12n4N12.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 63 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
5771344378527462685717311992153650392766866023261697486067156196504060456
990227796547638478624557875647494157966182026668594070587605661013239231
5175247374165152562934844040743858678921397329
```

denominator:

```
198412698412721821961606243750085083009806528493253321693578735752311484
753412029806539263278261149299761113327335196334201656942088814711445355
8910480033054593243558975630374221643754954960
```

B Coordinates

12 blocks of 12 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m12n4N12.txt`.

```
-393830647783 126873653371 102777653711 150115275396 198548539546 251147146916 -1970642535 -392035462938 41212585447 -26963871647 172324892002 -453368194391
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